Faculty of Computers and Information Technology

Department of Networks and Mobile

Graduation Project 2

❖ Course Information

Course Name: Graduation Project	**Course No:** ITNM4315
Prerequisite: -	**Credit Hours:**
Semester: Second 2025/2026	
Instructor: Hussam Musa. WhatsApp: 00972599291750, Email: hussam.s.musa@gmail.com	

❖ Description

This course is a complementary course allows students to apply the knowledge and skills acquired throughout the program to a real-world problem or research question within computer science networks and mobile. Students are supposed to go directly for coding with the following duties: implementation, results and reporting.

❖ Project Grade and Outline

Topic	Assessment Percentage	Expected Date
Chapter 4: Implementation and Development	60%	7-Jun-2026
Chapter 5: Results and Discussion	25%	7-Jul-2026
Chapter 6: Conclusion and Future Work **Chapter 7: References** **Chapter 8: Appendices** **Full Report**	15%	15-Jul-2026

❖ **More details will be discussed in lectures at these dates:**

Task	Date
Chapter 4: Implementation and Development **Chapter 5: Results and Discussion**	**6/Apr/2026 3pm**
Chapter 6: Conclusion and Future Work **Chapter 7: References** **Chapter 8: Appendices**	**4/May/2026 3pm**

Chapter 4: Implementation and Development

This chapter shows the actual work you did. It provides enough technical detail for someone to understand your implementation decisions. Expected length: 12–20 pages.

4.1 Development Environment Setup (1 page)

Document the complete setup process:

- Hardware specifications (processor, RAM, storage, network interfaces)
- Operating system and configuration
- Software installation steps and dependencies
- Network lab setup (for networking projects): switches, routers, virtual machines
- Simulation environment configuration (GNS3, Mininet, Packet Tracer)

4.2 System Architecture Implementation (2–3 pages)

Show how you translated the design from Chapter 3 into a working system:

- Final architecture diagram (may differ slightly from design)
- Component-by-component implementation details
- Communication interfaces and protocols implemented
- Configuration files and parameters

4.3 Core Functionality Implementation (5–8 pages)

This is the largest section. Break it into subsections by feature or module.

For Each Major Feature/Module, Include

1. Feature description: What does it do and why?
2. Algorithm or logic: Flowchart, pseudocode, or step-by-step explanation
3. Code snippets: Include the most important code with detailed comments
4. Design decisions: Why did you choose this approach?
5. Challenges and solutions: Problems encountered and how you solved them

> ⚠ **CODE SNIPPET GUIDELINES**
>
> Include ONLY the most important code — not entire files
>
> Every code snippet must have comments explaining what it does
>
> Use syntax highlighting or monospace font (Courier New)
>
> Explain the code in the text before and after the snippet

Reference line numbers when explaining specific parts

Put complete source code in the Appendix, not here

✎ **CS/NETWORKING IMPLEMENTATION EXAMPLES**

- Packet Capture Module: Show how you capture and parse network packets
- Feature Extraction: Show how you extract features from network flows
- ML Model Training: Show model architecture, hyperparameters, training loop
- Routing Configuration: Show router configuration commands and protocol setup
- API Endpoints: Show REST API implementation with request/response formats
- Database Operations: Show CRUD operations and query optimization
- Authentication: Show login, session management, access control implementation
- Encryption: Show data encryption/decryption implementation

4.4 User Interface Implementation (2–3 pages)

Show screenshots of every major screen in your application with explanations:

- Login/registration screens
- Main dashboard or home screen
- Key feature screens with data visualization
- Settings and configuration pages
- Alert/notification displays

For each screen: describe the purpose, key UI elements, and user interaction flow.

4.5 Security Implementation (1–2 pages)

Describe all security measures implemented:

- Authentication mechanism (password hashing, token-based auth, OAuth, etc.)
- Authorization and access control (RBAC, ACLs)
- Data encryption (at rest and in transit)
- Input validation and sanitization
- Network security measures (firewall rules, HTTPS, SSL/TLS configuration)

4.6 Network Configuration (for Networking Projects) (2–3 pages)

If your project involves network infrastructure:

- Show actual configuration commands (router, switch, firewall, server)
- Document VLAN, ACL, NAT, DHCP, DNS configurations

- Show network topology as implemented
- Include verification commands and their outputs (ping, traceroute, show commands)

4.7 Challenges and Solutions (1–2 pages)

Document major technical challenges encountered during implementation and how you resolved them. This shows problem-solving ability and adds credibility to your work.

Chapter 5: Results and Discussion

This chapter presents your testing results, analyzes them, and discusses their implications. Expected length: 10–15 pages.

5.1 Testing Environment (1 page)

Document the exact testing setup:

Component	Specification
Hardware	Intel Core i7-12700H, 16GB RAM, 512GB SSD, Gigabit Ethernet NIC
Operating System	Ubuntu 22.04 LTS (kernel 5.15)
Network Simulator	Mininet 2.3.1 / GNS3 2.2
Dataset	CICIDS2017 (2,830,743 records, 78 features, 14 attack types)
Testing Tools	Wireshark 4.0, iperf3, Apache JMeter 5.6
Statistical Analysis	Python 3.11, pandas, matplotlib, scipy

5.2 Unit and Integration Testing Results (2–3 pages)

Present results in structured tables:

Test ID	Test Description	Input	Expected Output	Actual Output	Status
UT-01	Validate packet parser	Sample PCAP file	Parsed header fields	Correctly parsed	PASS
UT-02	Test feature extraction	100 network flows	78 features per flow	78 features extracted	PASS
UT-03	Test ML prediction	Known attack sample	Attack label: DDoS	Predicted: DDoS	PASS
UT-04	Test DB logging	Alert event object	Record in database	Record stored	PASS
IT-01	End-to-end detection	Live traffic capture	Alert on attack	Alert generated	PASS

5.3 Performance Results (3–4 pages)

Present quantitative results using tables and charts. This is where your project proves its value.

Metrics to Report (Choose Based on Project Type)

Project Type	Key Metrics to Measure
ML/IDS Project	Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC/AUC, Processing Speed
Network Performance	Throughput, Latency, Jitter, Packet Loss, RTT, Bandwidth Utilization
Web Application	Response Time, Requests/Second, Error Rate, Page Load Time, Concurrent Users
Security Project	Vulnerability Count, Time to Detect, False Positive Rate, Encryption Overhead
IoT Project	Message Delivery Rate, End-to-End Delay, Energy Consumption, Reliability
SDN Project	Flow Setup Time, Controller Throughput, Failover Time, Rule Processing Rate

✓ PRESENTING RESULTS EFFECTIVELY

Use charts/graphs (line charts for trends, bar charts for comparisons, heatmaps for confusion matrices)

Always label axes, include units, and add titles to all figures

Include statistical measures: mean, standard deviation, confidence intervals

Present both successful and unsuccessful results — honesty adds credibility

Number all tables (Table 5.1, Table 5.2) and figures (Figure 5.1) and reference them in text

5.4 Comparative Analysis (2–3 pages)

Compare your system with existing solutions:

Metric	Your System	Snort	Suricata	Study X (2023)
Detection Accuracy	97.3%	89.2%	91.5%	94.1%
False Positive Rate	1.8%	5.2%	4.1%	3.5%
Processing Speed	12,000 pkt/s	15,000 pkt/s	18,000 pkt/s	8,000 pkt/s
Zero-Day Detection	Yes (anomaly)	No	Limited	Yes
Setup Complexity	Low (automated)	High	High	Medium

5.5 Discussion and Interpretation (2–3 pages)

Analyze and interpret your results:

6. What do the results mean? Explain the significance of each major finding.
7. Why did you get these results? Discuss the reasons behind your performance numbers.
8. How do results compare with literature? Confirm or challenge findings from Chapter 2.
9. Were objectives achieved? Map each result back to your objectives from Chapter 1.
10. What are the practical implications? How would these results affect real-world deployment?

5.6 Limitations of Results (1 page)

Be honest about limitations:

- Testing environment constraints (lab vs. real-world network)
- Dataset limitations (synthetic vs. real traffic, dataset age)
- Sample size or testing duration constraints
- Factors that could affect generalizability
- What additional testing would strengthen the results

Chapter 6: Conclusion and Future Work

Expected length: 4–6 pages.

6.1 Summary of Work (1–2 pages)

Provide a concise overview of the entire project: what problem you addressed, how you solved it, and what you found. Reference each chapter briefly.

6.2 Achievement of Objectives (1 page)

For each objective from Chapter 1, state clearly whether it was achieved and provide brief evidence:

Objective	Status	Evidence
Design hybrid IDS architecture	Fully Achieved	Six-component architecture implemented (Section 4.2)
Implement ML detection engine	Fully Achieved	RF + LSTM model trained and deployed (Section 4.3)
Train on CICIDS2017 dataset	Fully Achieved	2.8M records processed, 78 features (Section 4.3)
Evaluate accuracy and speed	Fully Achieved	97.3% accuracy, 12K pkt/s (Section 5.3)
Compare with Snort/Suricata	Fully Achieved	+8.1% accuracy over Snort (Section 5.4)

6.3 Key Contributions (0.5–1 page)

List 3–5 specific contributions your project makes:

11. Novel contribution: What is new about your work?
12. Technical contribution: What technical solution did you create?
13. Practical contribution: What practical tool or system did you deliver?
14. Knowledge contribution: What insights did your evaluation reveal?

6.4 Future Work (2–3 pages)

Short-term Improvements (3–6 months)

- Bug fixes, UI improvements, additional features within current scope
- Example: Add support for IPv6 traffic analysis, implement real-time alerting via email/SMS

Medium-term Extensions (6–18 months)

- Major feature additions, platform expansion, integration with other systems
- Example: Integrate with SIEM systems, develop mobile monitoring app, add automated response capabilities

Long-term Research Directions (1–3 years)

- Fundamental research questions, emerging technologies, large-scale deployment
- Example: Investigate federated learning for distributed IDS, explore quantum-resistant encryption, study adversarial ML attacks on IDS

6.5 Personal Reflection (0.5–1 page)

Reflect honestly on what you learned — technically, professionally, and personally. Mention what you would do differently if you could start over.

Chapter 7: References

Use IEEE citation style (standard for CS and networking). List all sources cited in the document.

IEEE Format Examples

Journal Article: [1] A. Author and B. Author, "Title of article," Journal Name, vol. X, no. Y, pp. ZZ-ZZ, Month Year.

Conference Paper: [2] C. Author, "Title of paper," in Proc. Conference Name, City, Country, Year, pp. ZZ-ZZ.

Book: [3] D. Author, Title of Book, Edition. City: Publisher, Year.

Website: [4] E. Author, "Page title," Website Name, Month Day, Year. [Online]. Available: https://url.com

RFC: [5] F. Author, "RFC Title," RFC XXXX, Month Year. [Online]. Available: https://www.rfc-editor.org/rfc/rfcXXXX

Source Type	Target %	Where to Find
Peer-reviewed journals	40–50%	IEEE Xplore, ACM Digital Library, ScienceDirect, Springer
Conference papers	20–30%	IEEE conferences, ACM SIGCOMM/CCS, USENIX
Books and book chapters	10–15%	University library, Springer, O'Reilly
Technical reports / RFCs	5–10%	IETF, NIST, vendor whitepapers
Reputable websites	5–10%	Official docs, GitHub repos, standards bodies

⚠ CRITICAL REFERENCE RULES

Every source cited in text MUST appear in the reference list (and vice versa)

80% of references should be from 2020–2025

Use reference management software: Zotero (free), Mendeley, or EndNote

Never fabricate references — this is academic fraud

Verify all URLs are accessible at time of submission

Chapter 8: Appendices

Include all supplementary material that supports the main text but would disrupt the flow if included in the chapters.

- **Appendix A: Complete Source Code:** Full code listings organized by module, well-commented. If code is very long, provide a GitHub repository link.
- **Appendix B: Network Configuration Files:** Complete router/switch/firewall configurations, VLAN setup files, ACL rules, server configuration files.
- **Appendix C: Complete Test Results:** Full test case documentation, complete test logs, raw performance data, all test screenshots.
- **Appendix D: System Diagrams:** Detailed UML diagrams (class, sequence, activity, use case), ER diagrams, network topology diagrams, flowcharts.
- **Appendix E: User Manual:** Step-by-step installation guide, user guide with screenshots, administrator guide, troubleshooting FAQ.
- **Appendix F: Surveys and Questionnaires:** Copies of all evaluation instruments used, consent forms, user testing forms.
- **Appendix G: Dataset Documentation:** Dataset description, preprocessing steps, feature definitions, sample data records.
- **Appendix H: Glossary:** Definitions of all technical terms, acronyms, and abbreviations used in the document.
- **Appendix I: Project Timeline:** Detailed Gantt chart, milestones, and deliverables achieved.